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safuzzy: an R package for adaptability and stability analysis with fuzzy inference

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Software

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Summary

Recommending genotypes for cultivation in a specific region is one of the main steps in genetic improvement programs. Several methodologies for genotype-by-environment interaction are available. However, they require careful evaluation based on indices and/or parameters, which can hinder selection when the number of genotypes increases. Fuzzy logic emerges as a valuable tool in the selection process, as it is capable of reflecting human thought. The safuzzy package was developed in R to implement phenotypic adaptability and stability analyses using fuzzy logic. With intuitive syntax, the package classifies genotypes by degrees of membership in recommendation groups, simplifying selection.

Stat of need

Modern plant breeding faces the constant challenge of developing genotypes that not only exhibit high yield potential but also maintain performance stability across a wide range of environmental conditions. The recommendation of cultivars depends on methodologies capable of understanding genotype-environmental interactions. Among the tools available to breeders, adaptability and stability analyses play a fundamental role in quantifying phenotypic behavior in response to environmental fluctuations, allowing the identification of materials with specific responses or broad adaptation.

Historically, regression-based methods proposed by Eberhart & Russell (1966) and Cruz et al. (1989), relative performance Lin & Binns (1988), or recommendation confidence index (Annicchiarico, 1992), are widely used in breeding programs. The methodologies of Lin & Binns (1988) and Annicchiarico (1992) represent the concept of adaptability and stability in a single parameter (Schmildt et al., 2011). However, both methods were subsequently modified to represent indices for favorable and unfavorable environments, and, as with regression-based methodologies, require the breeder to synthesize data to recommend the best genotype.

This complexity increases as a greater number of genotypes are added to cultivation and use value trials. Breeders frequently encounter genotypes with high average and moderate adaptability and/or stability coefficients, making a binary selection decision difficult. This complexity can represent a challenge for researchers and professionals in the field, especially when dealing with large volumes of data and the simultaneous application of different methods (Silva Júnior et al., 2021). In this context, tools that integrate methodologies in a unified and automated environment are highly valuable for streamlining the analytical process and improving the reproducibility of analyses.

Fuzzy logic has emerged as a promising approach to support decision-making in breeding programs, as it allows not only the classification of genotypes into adaptability and stability groups, but also the quantification of the degree to which each genotype belongs to these groups. This characteristic is particularly useful because it reflects the reasoning used by

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experts, making the interpretation of results more intuitive (Maciel et al., 2025,).

Despite the proven effectiveness of fuzzy logic in adaptability and stability studies, its implementation is still fragmented, requiring extensive computational routines for parameter extraction, definition of membership functions, construction of linguistic rules, and inference. The `safuzzy` package (Stability Adaptability with Fuzzy Logic) was developed to address this need. It offers a user-friendly implementation in R (R Core Team, 2025) of classical methods with fuzzy inference.

State of the field

While existing R packages implement classical individual adaptability and stability analyses, `safuzzy` is the first to integrate these traditional methods with fuzzy logic inference systems. The package implements five core functions corresponding to the methodologies described in the literature:

- **Eberhart & Russell with Fuzzy Inference** (`eberhart_russell()`): Implements the regression-based model combined with fuzzy logic inference as proposed by V. Q. Carneiro et al. (2018).
- **Lin & Binns Superiority Index with Fuzzy Inference** (`lin_binns()`): Evaluates non-parametric superiority measures integrated with fuzzy rules based on the interpretation of the methodology.
- **Annicchiarico Recommendation Index with Fuzzy Inference** (`annicchiarico()`): Calculates confidence indices with fuzzy inference systems following A. R. T. Carneiro et al. (2019).
- **Cruz, Torres & Vencovsky Bi-Segmented Model with Fuzzy Inference** (`cruz_torres_vencovsky()`): Implements bi-segmented non-linear regression models coupled with fuzzy controllers as described by A. R. T. Carneiro et al. (2019).
- **Hybrid Controller** (`hybrid()`): Combines non-parametric superiority indices and regression parameters into a single fuzzy controller propused by A. R. T. Carneiro et al. (2020).

Software Design

The `safuzzy` package was designed following modern R programming practices and the architectural principles of the `tidyverse` ecosystem. All core functions accept standard data frames and leverage `dplyr` for non-standard evaluation and internal data transformations, consistently returning standardized `tibble` structures.

The software architecture decouples the statistical parameter extraction step from the fuzzy inference mechanics. Each classical model (`eberhart_russell`, `lin_binns`, `annicchiarico`, `cruz_torres_vencovsky`) and the combined `hybrid` controller operates as an independent, modular pipeline. Membership functions (`zmf`, `smf`, `pimf`) and standardization scaling are computed via vectorised internal helper functions. Function documentation and parameter descriptions are generated using `roxygen2`, and unit testing is implemented via `testthat` to guarantee stability and continuous integration.

safuzzy is available on CRAN.

Usage Example

The following example demonstrates how to perform a fuzzy adaptability and stability analysis using the Eberhart & Russell methodology on a multi-environment dataset:

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```r
library(safuzzy)

# Load example dataset containing Genotypes, Environments, Replications and Yield
library(agridat)
data <- gauch.soy

# Run Eberhart & Russell analysis with fuzzy logic
results <- eberhart_russell(
  data = data,
  gen = gen,
  rep = rep,
  env = env,
  var = yield
)

# View summary output table
print(results)
```

AI usage disclosure

Generative AI tools (Gemini flash) were utilized during the development of this project to assist in code and polishing the English translation. All generated code, and text were reviewed, and verified by the authors.

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